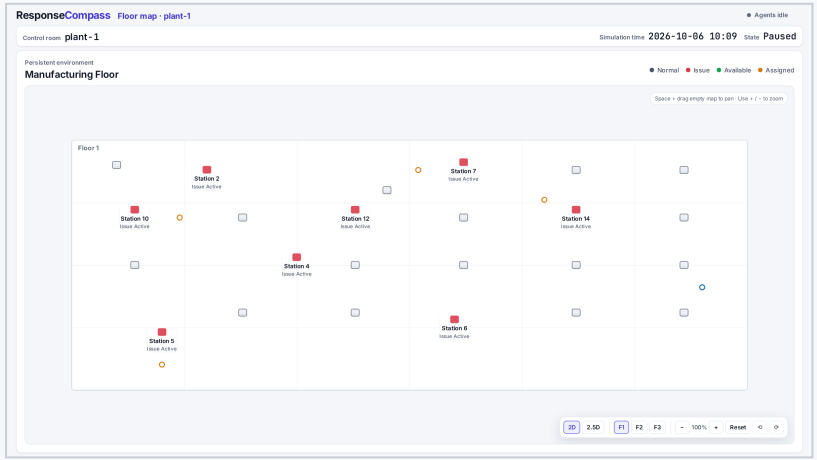
ResponseCompass

The right responder. The right fix.

Learned from every past repair.

Runs autonomously. Supervised by your control room.



Floor-map wall screen, simulated plant e8-aura-p1

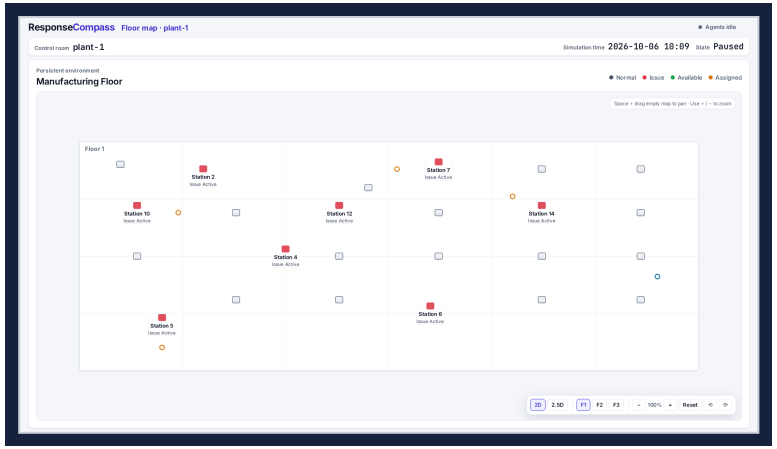
An AI dispatch agent for operations control rooms, built on AMD Token Factory.

The problem

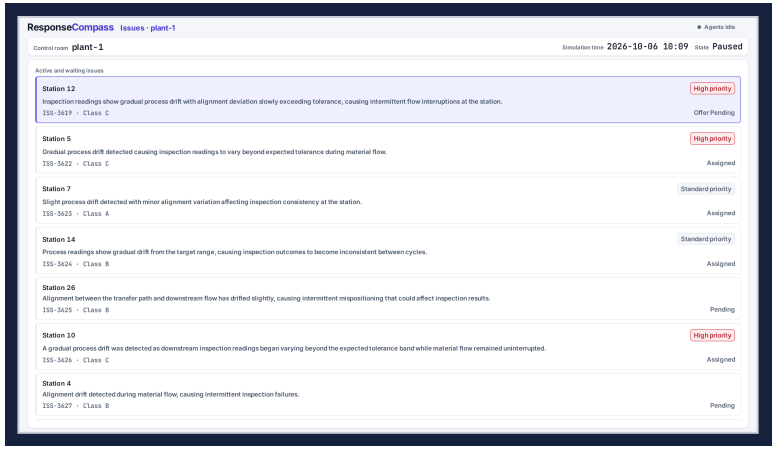
- When equipment fails, **closest** or **free** may be an incomplete answer.
- The nearest technician may never have fixed this kind of fault, and the first fix tried is often a guess.
- A poor match means downtime, a reopened issue or another trip; a backlog means overtime decisions under pressure.
- Who fixes what, and how, lives in people's heads and old tickets.

One agent, many screens

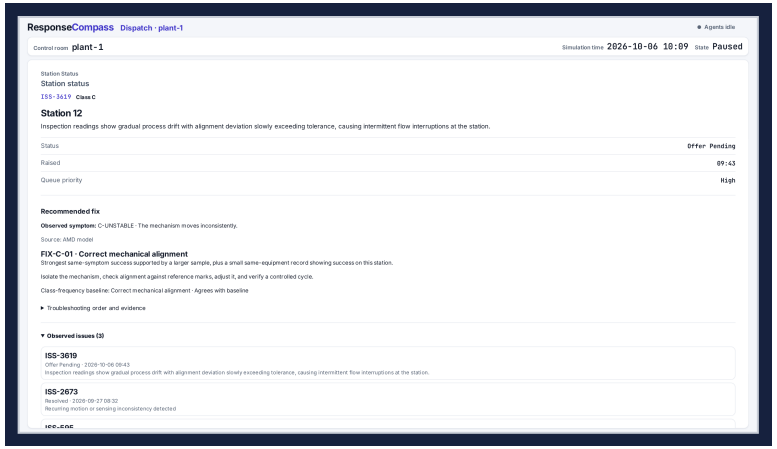
Each panel can run full screen on its own monitor; the operator console keeps the controls.



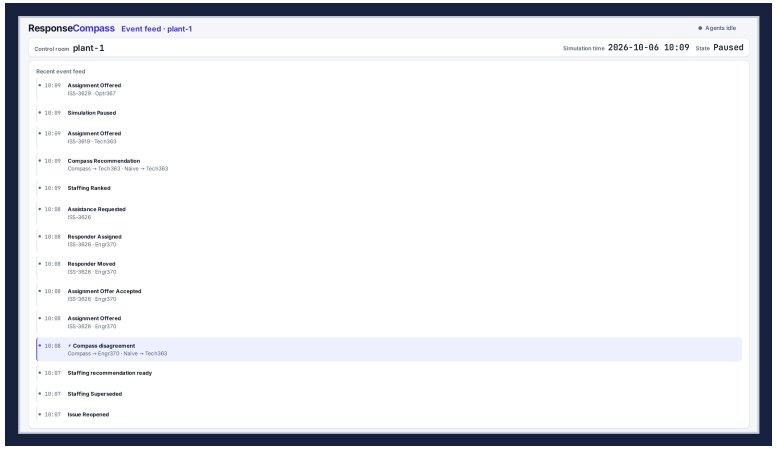
Floor map



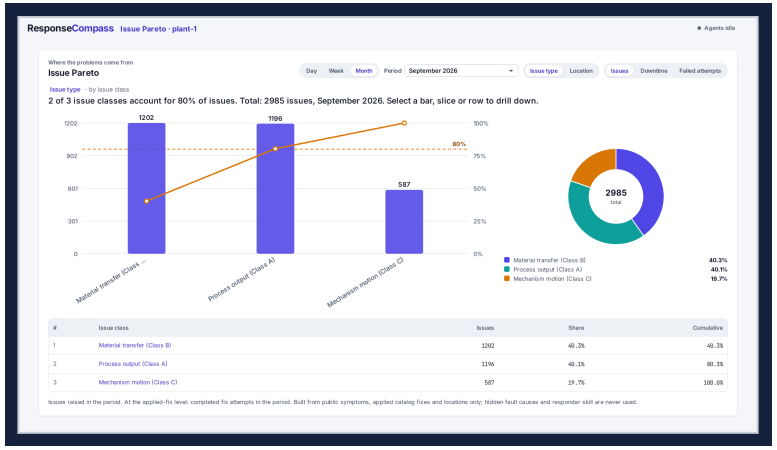
Issues



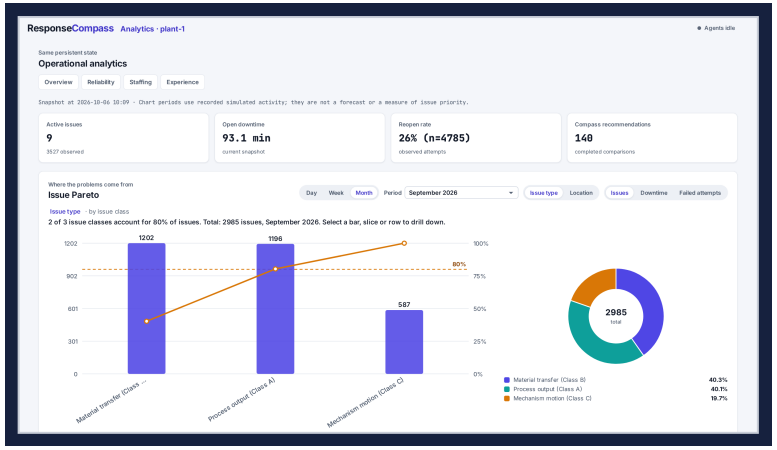
Dispatch



Event feed



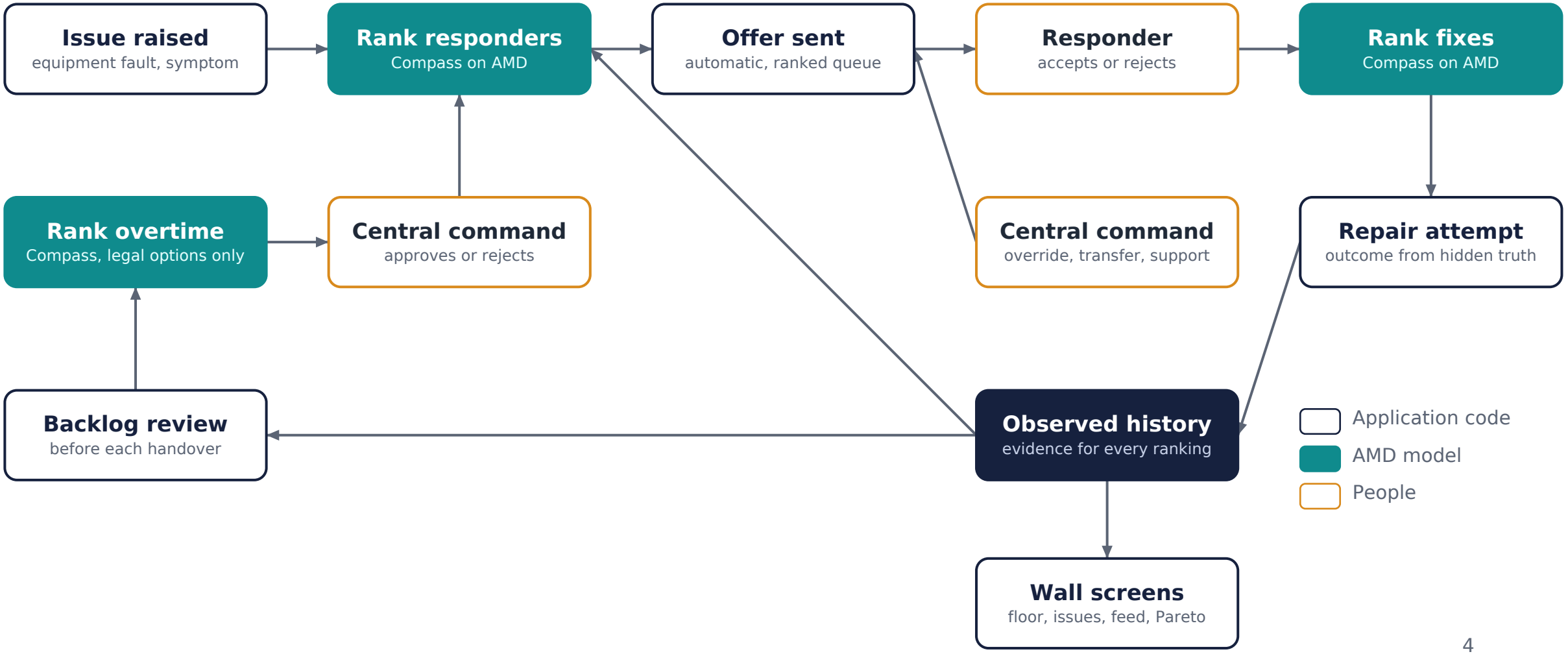
Issue Pareto



Analytics

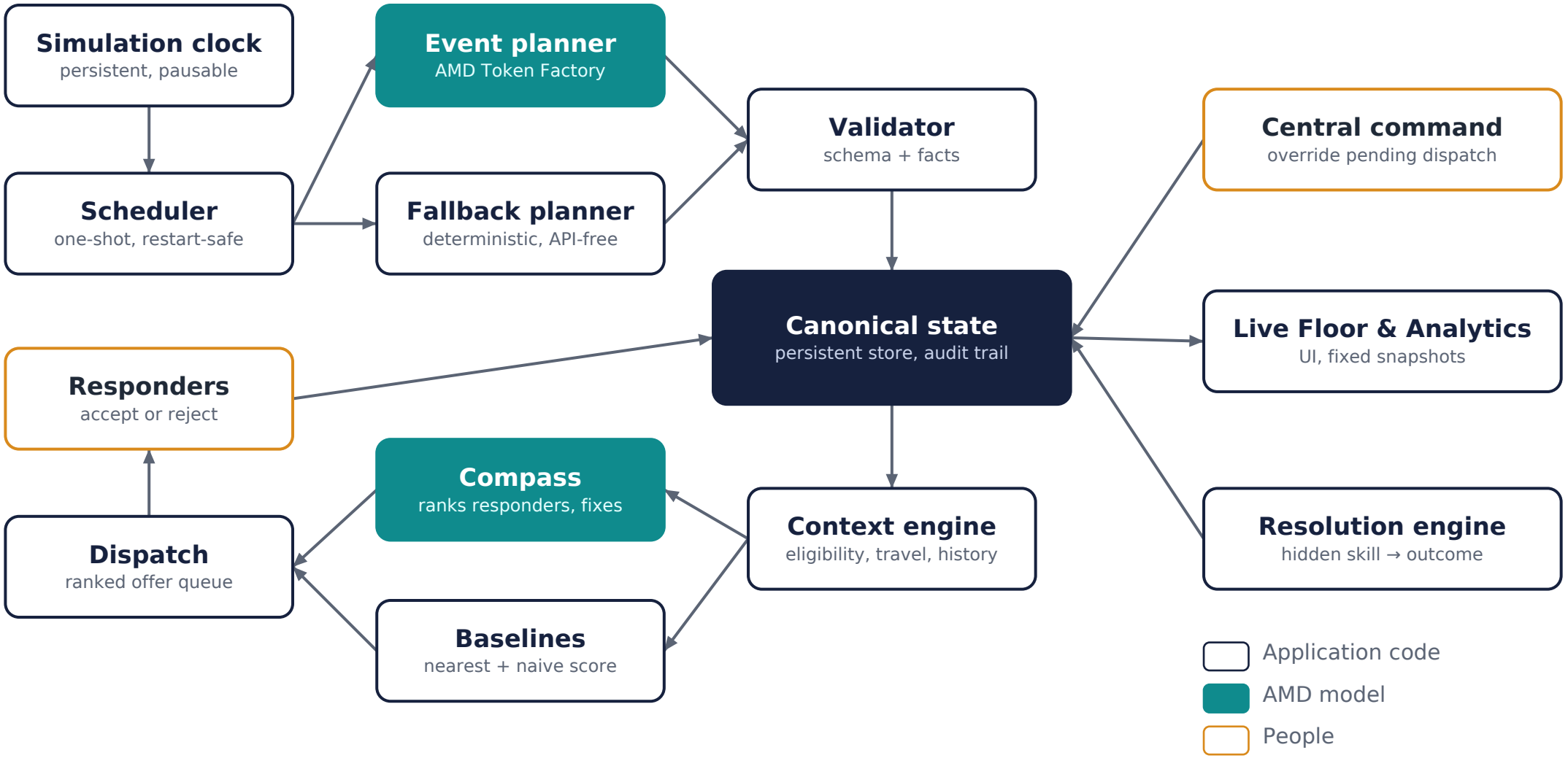
Autonomous by default, human in command

Code runs the loop; AMD models rank; people decide at the amber steps.



Code owns the facts. Models only rank.

Application code owns time, facts, eligibility, baselines and every mutation; models only propose.



Honest by design

- Each responder's **true skill** and each fault's **true cause** are hidden simulation truth: no model, screen or analytics sees them.
- Outcomes are simulated from that hidden truth, independently of what the model claims.
- Transparent baselines run alongside (nearest, naive score, most common fix); disagreements are recorded.
- Analytics show denominators, suppress thin samples and are not an employee leaderboard.
- Excluded from ranking: pay, seniority, demographics, fatigue, HR data.

The right responder

ISS-3618, Station 48 · captured live on AMD Token Factory (MiMo-V2.6-Flash)

- Nearest responder **Engr361**: half a minute away, 26 similar cases.
- Compass ranked **Engr370** first: 110 similar cases, 87% observed success, the same half-minute away, and explained why.
- Offer accepted automatically; central command could have overridden it at any time.

Initial assignment Compass analysis

Engr370 was ranked first.

Engr370 combines the deepest issue-specific history, the most reliable evidence, and the best observed balance of success and limited reopens, while still requiring only a modest trip relative to the alternatives.

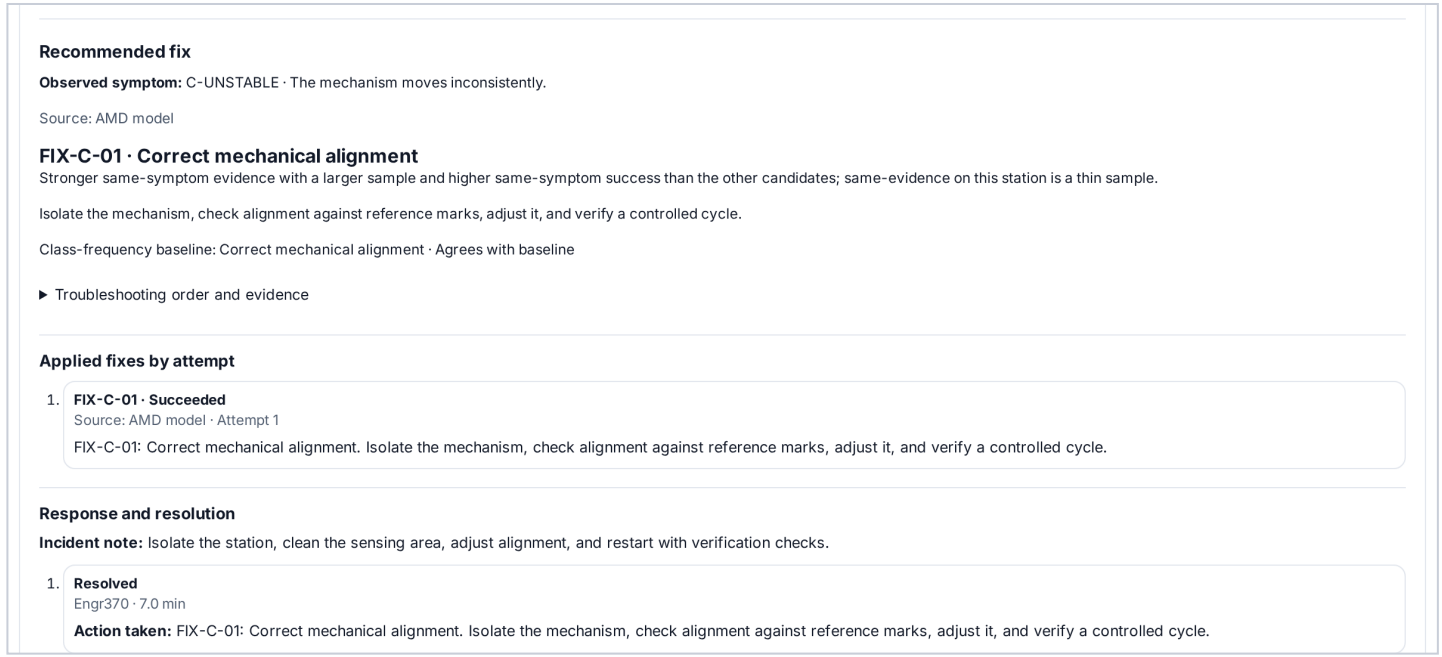
Eligible responder evidence

Candidate	Solo / assisted	Evidence weight	Strength	Median	Success	Travel
1. Engr370	17 / 93	62.9	STRONG	8.8 min	87% (n=62.9)	0.5 min
2. Engr361	3 / 23	10.4	STRONG	13.9 min	83% (n=10.4)	0.5 min
3. Tech374	0 / 7	4.1	MODERATE	12.1 min	66% (n=4.1)	1.8 min

The right fix

Same record: Compass ranks the troubleshooting procedures for the observed symptom

- Code supplies the catalog procedures and their **observed** success on this symptom, class and station.
- Compass ranked **Correct mechanical alignment** first; the most common fix is the baseline.
- Engr370 applied it: fixed **first time in 7.0 min**.
- A failed attempt reopens the issue and the next procedure is tried.



Recommended fix

Observed symptom: C-UNSTABLE · The mechanism moves inconsistently.

Source: AMD model

FIX-C-01 · Correct mechanical alignment

Stronger same-symptom evidence with a larger sample and higher same-symptom success than the other candidates; same-evidence on this station is a thin sample.

Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Class-frequency baseline: Correct mechanical alignment · Agrees with baseline

▸ Troubleshooting order and evidence

Applied fixes by attempt

1. **FIX-C-01 · Succeeded**
   Source: AMD model · Attempt 1
   FIX-C-01: Correct mechanical alignment. Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Response and resolution

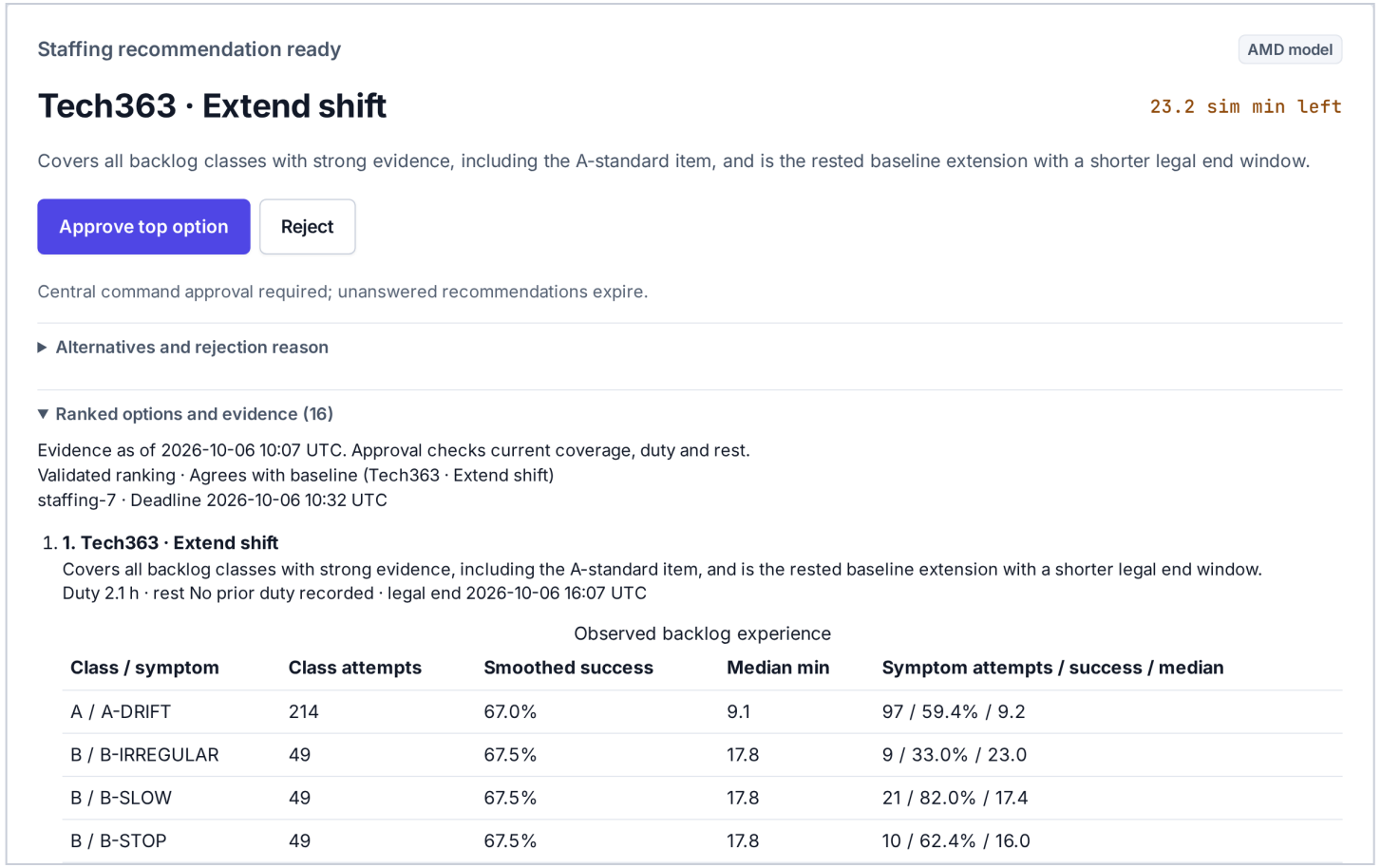
Incident note: Isolate the station, clean the sensing area, adjust alignment, and restart with verification checks.

1. **Resolved**
   Engr370 · 7.0 min
   Action taken: FIX-C-01: Correct mechanical alignment. Isolate the mechanism, check alignment against reference marks, adjust it, and verify a controlled cycle.

Overtime, decided by people

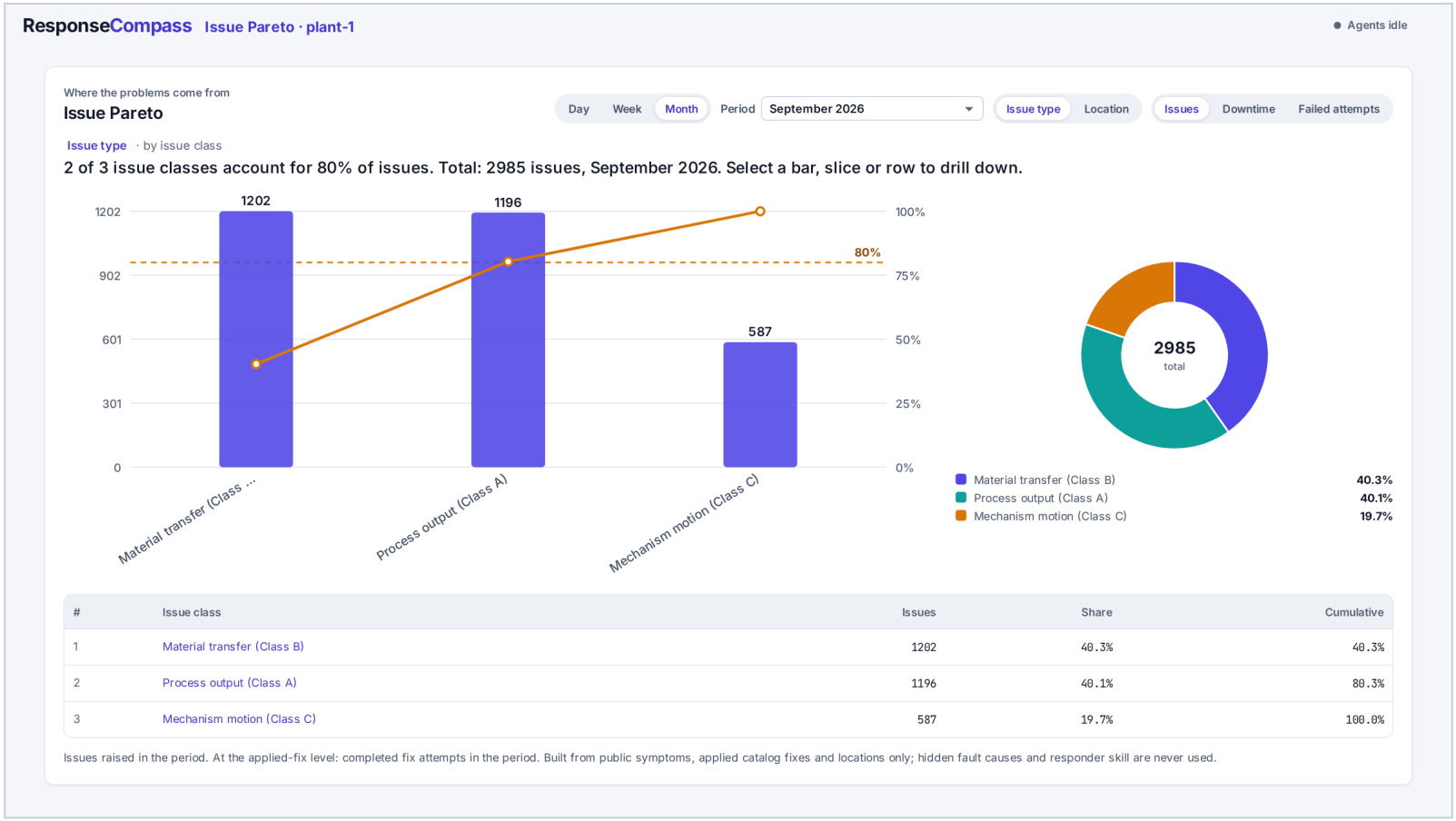
Backlog review → lawful options → Compass ranking → central command approves or rejects

- When the backlog outgrows the shift, code lists only **legal** options: 16 h duty, 12 h rest, staffing cap.
- Compass ranks them by observed coverage of the backlog; the old first-eligible rule is the baseline.
- Nothing changes until **central command** approves; unanswered recommendations expire.
- Every decision, and who made it, is audited.



Where the downtime comes from

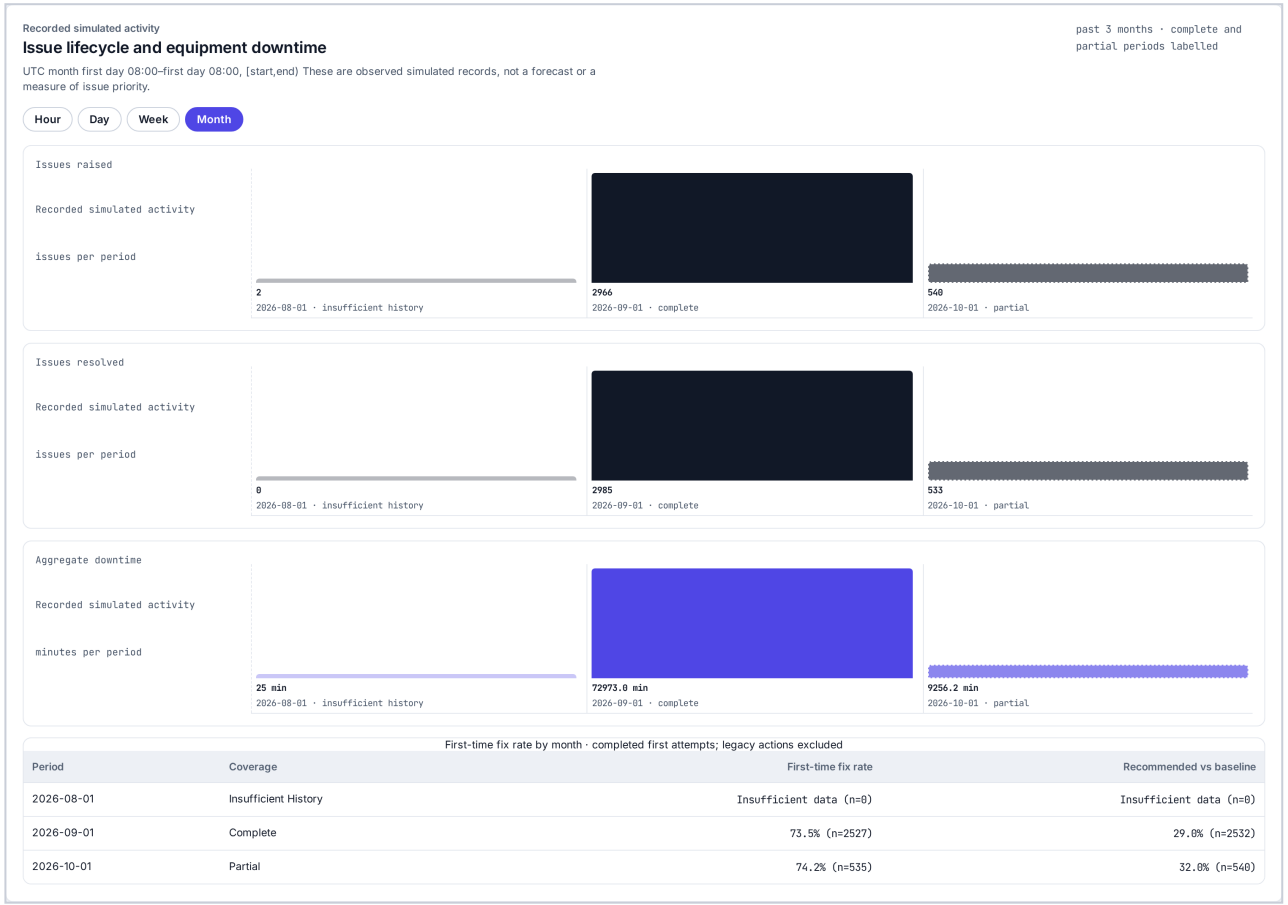
Issue Pareto: issue class → observed symptom → applied fix, or floor → station



- Issues, downtime or failed attempts per day, week or month.
- Sorted bars, cumulative line and an **80%** marker, plus a donut.
- Click to drill from class to symptom to the fix that was applied.
- Built from public records only; hidden causes are never used.

A month of operations

- e8-aura-p1: **3,512 issues**, 1 September to 6 October (simulated).
- September: **2,966 raised / 2,985 resolved** events.
- First-time fix rate in September: **73.5%** (1,858 of 2,527 first attempts).
- Weekly first-time fix: 67.1%, 74.9%, 76.5%, 75.6%: not a steady climb.

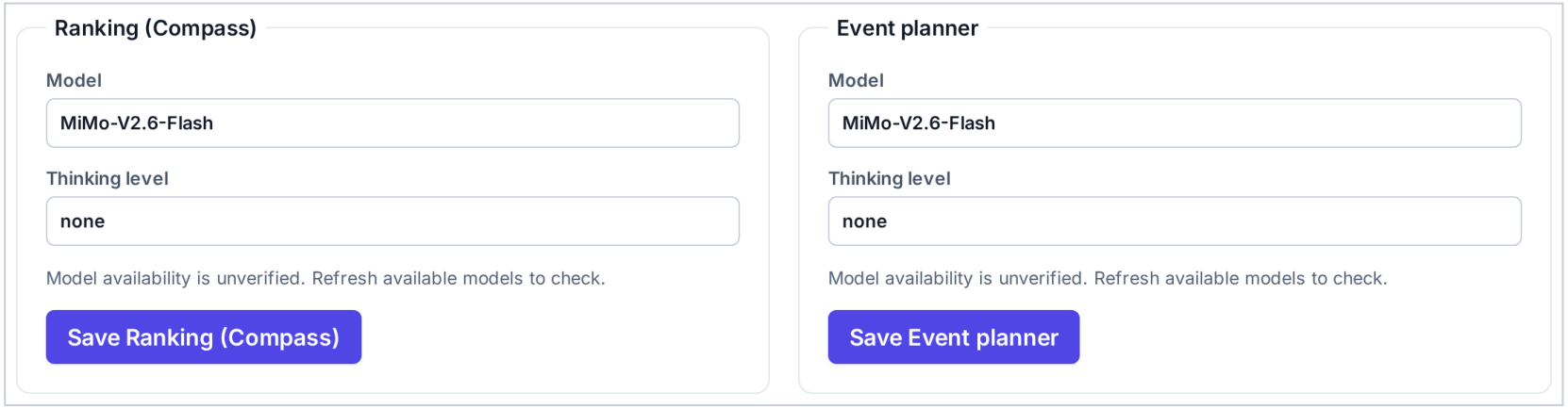


First-time fix rate by month · completed first attempts; legacy actions excluded

Period	Coverage	First-time fix rate	Recommended vs baseline
2026-08-01	Insufficient History	Insufficient data (n=0)	Insufficient data (n=0)
2026-09-01	Complete	73.5% (n=2527)	29.0% (n=2532)
2026-10-01	Partial	74.2% (n=535)	32.0% (n=540)

Built on AMD Token Factory

- OpenAI-compatible API, JSON output; our validators are the contract.
- Default model **MiMo-V2.6-Flash** ranks responders, fixes and overtime; the event planner can use another model.
- Local budget: 10 requests/min, 2 concurrent; bounded 429 retries; circuit breaker.
- Recorded MiMo check: **9/10** valid responder rankings, **10/10** valid event plans.
- Model calls run in a background lane and never hold the simulation clock.



Engineering quality

- Live and Offline are one app; Offline makes no API calls and is fully reproducible.
- **319** hermetic unit and integration tests and **71** Chromium end-to-end tests.
- Deterministic seeded simulation; restart-safe scheduler; atomic, crash-safe persistence.
- Runs on Node.js 22 or Docker; no external services besides the model API.

What's next

- Connect to real maintenance, ticketing and rostering systems in place of the simulated floor.
- Learn from approvals: feed operator decisions and repair outcomes back into the evidence.
- Measure Compass against its baselines on live AMD rankings over a longer run.
- Long-run storage: compact the remaining growing history for year-long runs.

The right responder. The right fix.